Batangas State University
Integrated School

SCIENCE 12-D
INTRODUCTION TO BIOTECHNOLOGY

The Biotechnology Century and Its Workforce

Module 1

Expected Learning Outcomes

1.1. Define biotechnology and understand the many scientific disciplines that contribute to biotechnology.

1.2. Provide examples of historic and current applications of biotechnology and its products.

1.3. List and describe different types of biotechnology and their applications.

1.4. Provide examples of potential advances in biotechnology.

The Big Idea!

Have you ever eaten corn chips, yogurt, cheese, or even had a glass of milk? If so, you've already experienced the impact of biotechnology in your everyday life! Many of the foods we enjoy today are created or improved through biotechnology — the use of living organisms to produce useful products. In fact, scientists can now modify certain crops and organisms to make them grow faster, resist pests, or even become more nutritious. These are called genetically modified (GM) foods, and while they have brought many benefits, they have also sparked debates about safety, ethics, and environmental effects.

Biotechnology is not just about food. It is a broad field that connects biology, chemistry, genetics, and technology to improve health, agriculture, and the environment. From developing life-saving medicines and vaccines to creating eco-friendly fuels, biotechnology continues to shape our world in remarkable ways.

In this module, you will explore what biotechnology really means and discover the different areas it touches. You will learn about the science behind it, the tools and techniques used by researchers, and the exciting careers that biotechnology offers. Understanding these basics will help you appreciate how this field continues to change our lives — and how it might inspire you to be part of its future.

Deepen Your Understanding!

Since the completion of the **Human Genome Project** in 2001, biotechnology has made remarkable progress. Scientists can now analyze entire genomes, leading to new discoveries that help diagnose and treat diseases more effectively. Through **genetic engineering**, genes can be combined or modified to create medicines that were once difficult to produce. For example, some protein-based drugs can now be absorbed through the skin with the help of nanovesicles, which are tiny delivery systems designed to release medicines at the right time.

Biotechnology is not only transforming healthcare but also improving energy and the environment. Certain bacteria are being used to produce

biofuels, offering a cleaner and renewable source of energy. Researchers are also creating bacteria with synthetic genomes to perform specific tasks, such as **cleaning up pollutants**. These efforts have helped advance **bioremediation**, which uses living organisms to remove harmful substances from the environment, such as the bacteria that helped degrade oil during the 2010 Gulf of Mexico spill.

In agriculture, the collaboration between nonprofit organizations and biotechnology companies has made it possible to develop **transgenic plants** that grow faster, resist diseases, and help reduce hunger. Similarly, advances in **vaccine development** have lowered healthcare costs for both humans and animals. DNA technology now also plays a major role in identifying individuals for medical, legal, and security purposes.

As we continue to explore the potential of biotechnology, aquatic biotechnology is becoming increasingly important. Scientists are discovering new medicines and materials from marine organisms that may lead to innovative treatments. While these advancements bring great promise, they also raise important ethical questions. It is essential to continue open discussions to ensure that biotechnology is used safely, responsibly, and for the good of all.

WHAT IS BIOTECHNOLOGY AND WHAT DOES IT MEAN TO YOU?

Biotechnology can be described as the use of living organisms or their products to improve life and solve problems. This means using parts of living things, such as cells, enzymes, or genes, to make useful products for people, animals, and the environment. Scientists and researchers in biotechnology are working toward a world where diseases are easier to cure, food is plentiful, and pollution can be reduced or removed.

Although biotechnology may seem new, its roots go back to **ancient times**. Early humans used **simple forms of biotechnology** when they made wine, baked bread, and preserved food with the help of microorganisms. What makes modern biotechnology different is the use of advanced tools like genetics, molecular biology, and computers to study and modify living systems at the microscopic level. These modern advances allow scientists to create new medicines, improve crops, and protect the environment more effectively.

We live in a time often called the "**century of biotechnology**" because of its growing importance and influence on human life. As you explore this field, you will learn not only how biotechnology works but also how it connects science and society. The knowledge you gain in this chapter will help you understand how this exciting science can make the world a healthier, safer, and more sustainable place for everyone.

A BRIEF HISTORY OF BIOTECHNOLOGY

When people hear the word biotechnology, they often imagine scientists in white lab coats doing secret experiments with complex machines. While that image shows part of the story, biotechnology is actually much older and more familiar than many think. In fact, humans have been practicing biotechnology for thousands of years. Ancient civilizations such as the Chinese, Greeks, Romans, Babylonians, and Egyptians used early forms of biotechnology as far back as 2000 B.C., long before the invention of modern laboratories.

One of the earliest examples of biotechnology is the **domestication of animals** like sheep, cows, and goats for food and clothing. Another ancient practice is **fermentation**, a process where microorganisms such as yeast and bacteria are used to produce foods and drinks. For instance, when baking bread, the yeast Saccharomyces cerevisiae ferments sugar and releases carbon dioxide, which causes the dough to rise. The same microorganism is also used to make wine and beer. These traditional practices show that even without advanced tools, early humans were already using living organisms to make useful products.

Over time, people learned that by selectively breeding plants and animals, they could improve their quality and productivity. Farmers crossbred crops that produced larger and sweeter fruits, while breeders raised animals that provided more meat or milk. This process allowed humans to take advantage of the genetic potential of organisms, even before the discovery of DNA. **Modern biotechnology** builds on these same ideas, but now scientists can **study** and **modify** genes directly using laboratory techniques.

A major turning point in biotechnology came in 1928, when **Alexander Fleming** discovered that a type of mold, **Penicillium**, could stop the growth of harmful bacteria. This led to the discovery of **penicillin**, the **first antibiotic**, which has saved millions of lives. During the 1940s and 1950s, scientists developed ways to grow bacteria in large amounts to produce antibiotics and other useful substances. This was the beginning of large-scale or "batch" production, allowing biotechnological products to be made efficiently and commercially.

By the 1960s and 1970s, advances in genetics and molecular biology transformed biotechnology into a modern science. Scientists discovered the structure of DNA and developed techniques to clone and manipulate genes. The ability to combine DNA from different organisms, known as **recombinant DNA technology**, made it possible to produce important substances such as **insulin**, **human growth hormone**, and **blood-clotting proteins**. This breakthrough marked the start of modern biotechnology as an industry.

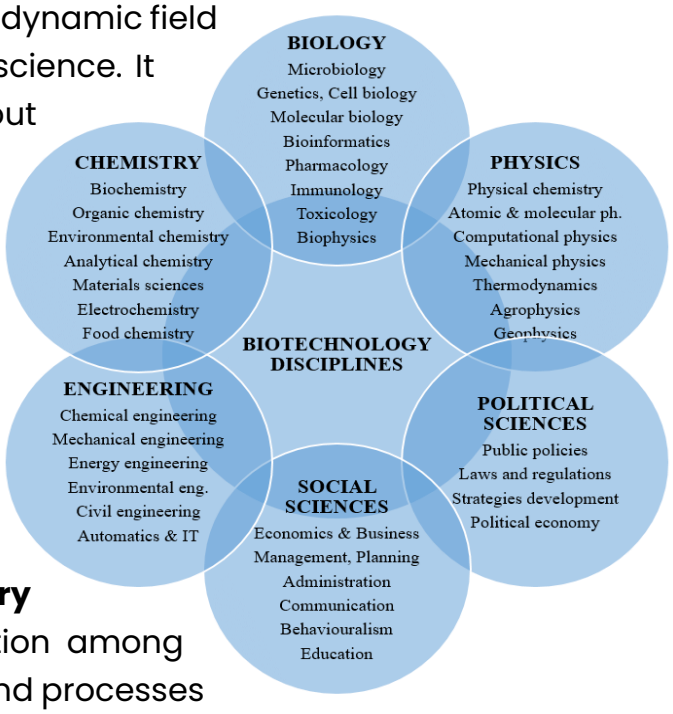
The most ambitious genetic research effort, the **Human Genome Project**, began in 1990. Its goal was to identify and map all the genes in human DNA. Completed in 2003, the project gave scientists a complete picture of the human genetic code, paving the way for new diagnostic tools and treatments for many diseases. It also opened new fields of research in medicine, genetics, and molecular biology. The 1000 Genomes Project followed, allowing scientists to compare human genes with those of other organisms to better understand how genes work.

From ancient fermentation to the mapping of the human genome, biotechnology has evolved into one of the most important and exciting fields of science. It connects the wisdom of the past with the discoveries of the present, shaping the future of medicine, agriculture, and the

environment. As you continue studying this field, you will encounter many new terms and ideas, but remember that biotechnology has always been driven by a simple goal: to use living things to improve life on Earth.

BIOTECHNOLOGY: A SCIENCE OF MANY DISCIPLINES

Biotechnology is a broad and dynamic field that **connects** many branches of science. It cannot be fully understood without recognizing the important roles played by **biology**, **chemistry**, **mathematics**, **computer science**, and **engineering**, along with other areas like **philosophy** and **economics**. Although much of biotechnology focuses on molecular biology techniques, it is far from a single, narrow discipline. Instead, it is an **interdisciplinary science** that thrives on collaboration among different fields to create products and processes that improve life and the environment. This combination of knowledge and skills from diverse areas also opens up a wide range of career opportunities for people with various scientific backgrounds.

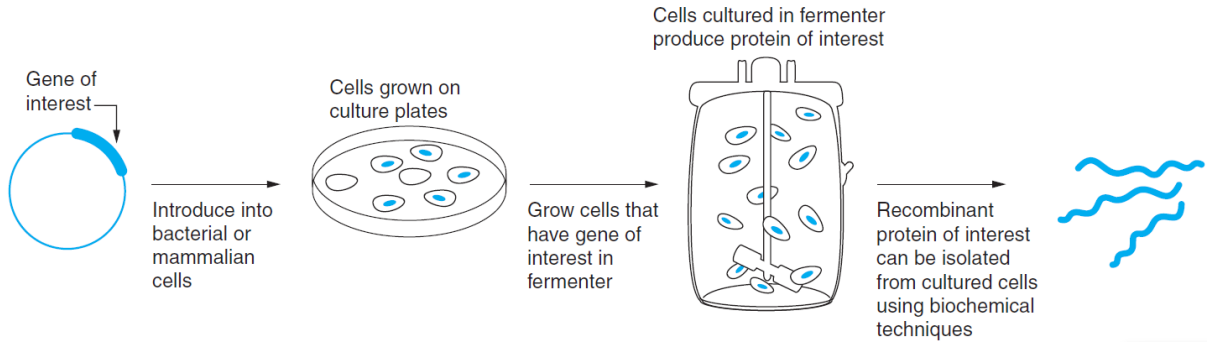


At its foundation, biotechnology is rooted in **basic sciences** that study living organisms at biochemical, molecular, and genetic levels. These "roots" form the basis of genetic engineering, supported by computer science tools that allow scientists to analyze genetic data and design biological solutions. For example, researchers in microbiology may discover a bacterial gene that could help treat a disease. Through **biochemical** and **genetic analysis**, and with the aid of **bioinformatics**, scientists determine how this gene works and how its protein product functions. Once understood, the gene may be used in **drug development**, **agriculture**, or **environmental management**. This example highlights how biotechnology integrates different disciplines to transform discoveries into real-world applications, showing that it is

indeed a science of many disciplines working together toward innovation and human advancement.

PRODUCTS OF MODERN BIOTECHNOLOGY

Modern biotechnology has produced an incredible range of innovative products that benefit humans, animals, and the environment. These advancements span various scientific fields, including microbiology, marine biology, and plant biology. Among the most significant contributions of biotechnology are products developed for human health, particularly in pharmaceutical production. In fact, more than 65 percent of biotechnology companies in the United States focus on creating drugs to treat different diseases. A milestone in this field was reached in 1982 when **Genentech**, recognized as the first biotechnology company, introduced recombinant insulin for diabetes treatment, marking the first biotechnology product approved for human use. Today, there are hundreds of biotech drugs, vaccines, and diagnostic tools available, with more than 300 additional medicines currently in development to target over 200 diseases, especially cancer, which remains a major focus of biotech research.



A large number of biotechnology products are **recombinant proteins**, which are proteins produced through gene-cloning techniques. This process involves inserting human genes into bacterial or mammalian cells to produce specific proteins that can be used to treat diseases such as **AIDS**, **diabetes**, and **cancer**. The cells used for this purpose are grown in a process known as cell culture, where they

TABLE 1 TOP 10 BIOTECHNOLOGY DRUGS (WITH SALES OVER $1 BILLION)

Drug	Developer	Function (Treatment of Human Disease Conditions)
Enbrel	Amgen & Wyeth	Rheumatoid arthritis
Remicade	Johnson & Johnson	Rheumatoid arthritis
Rituxan	Roche	Non-Hodgkin's lymphoma
Avastin	Roche	Colon cancer
Herceptin	Roche	Breast cancer
Humira	Abbott Labs	Rheumatoid arthritis
Levenox	sanofi-aventis	Blood clots
Lantus	sanofi-aventis	Diabetes
Aranesp	Amgen	Anemia

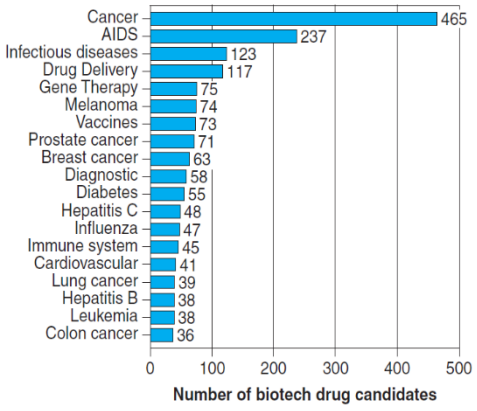
receive the nutrients needed for growth in controlled environments. To produce these proteins on a large scale, scientists use fermenters or bioreactors, which allow the mass cultivation of genetically modified cells. Once the target protein is produced, it is harvested and purified for medical or industrial use. This method of producing **life-saving treatments** demonstrates how biotechnology harnesses cellular processes to meet global health needs and improve the quality of human life.

TABLE 2 EXAMPLES OF PROTEINS MANUFACTURED FROM CLONED GENES

Product	Application
Blood factor VIII (clotting factor)	Treat hemophilia
Epidermal growth factor	Stimulate antibody production in patients with immune system disorders
Growth hormone	Correct pituitary deficiencies and short stature in humans; other forms are used in cows to increase milk production
Insulin	Treat diabetes
Interferons	Treat cancer and viral infections
Interleukins	Treat cancer and stimulate antibody production
Monoclonal antibodies	Diagnose and treat a variety of diseases including arthritis and cancer
Tissue plasminogen activator	Treat heart attacks and stroke

Biotechnology continues to transform human health through remarkable and promising innovations. One of the most groundbreaking developments is **gene therapy**, which involves introducing new genes into human cells to treat or even cure genetic diseases. In the near future, this approach may become a routine medical practice, offering hope for individuals suffering from inherited or otherwise untreatable conditions. Alongside this, genetics and tissue engineering are paving the way toward growing replacement organs for **transplantation**. These **bioengineered organs** are expected to have a lower risk of rejection, addressing one of the major challenges in organ transplantation today.

Marine biotechnology also plays a crucial role in medical advancement. Compounds derived from marine organisms are now being developed into treatments for **cancers**, **strokes**, and **arthritis**, highlighting the ocean's potential as a source of novel therapeutics. Additionally, the production of specialized proteins through transgenic animals, such as the clotting protein **ATryn** produced by genetically modified goats, has enabled the large-scale creation of vital medicines. These innovations, powered by discoveries from the Human Genome Project, are revolutionizing medicine by allowing more personalized and effective treatments. Ultimately, biotechnology not only enhances



the quality of human life but also holds the promise of extending life expectancy through continuous scientific progress.

ETHICS AND BIOTECHNOLOGY

Biotechnology offers remarkable benefits, but it also brings with it serious **ethical**, **legal**, and **social concerns** that continue to spark global debate. As scientific advances grow more powerful, questions arise about how far humans should go in altering life. Many people, including scientists, lawmakers, religious leaders, and the general public, share differing views on how biotechnology should be used responsibly. The discussion extends beyond the laboratory and affects everyone, especially as new technologies increasingly influence daily life and human health.

One major ethical issue is **cloning**. After the successful cloning of mammals such as sheep, cows, and monkeys, some have proposed that cloning could be applied to humans as well. This raises moral questions about identity, individuality, and the value of human life. Similarly, the **creation of synthetic genomes** and **proteins** introduces new controversies about how much humans should manipulate living systems. These issues show that biotechnology is not only a scientific field but also a moral and social one. Anyone pursuing a career in this discipline must learn to work with people who hold diverse perspectives and values. The goal is not to be told what to think, but to become informed enough to make ethical and thoughtful decisions about biotechnology's role in shaping the future.

TYPES OF BIOTECHNOLOGY

Biotechnology is a vast field with many branches, each contributing to the betterment of society, human health, and the environment. The major types include **microbial**, **agricultural**, **animal**, **forensic**, **bioremediation**, **aquatic**, and **medical biotechnology**. Each type applies biological knowledge and technology in unique ways to solve problems and improve life on Earth.

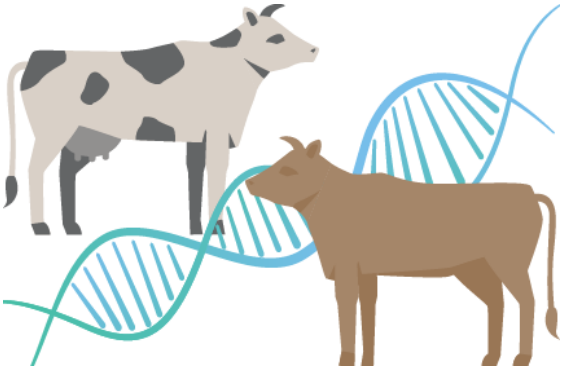
Microbial Biotechnology

Microbial biotechnology focuses on the use of **microorganisms** such as **bacteria**, **yeast**, and **fungi** to develop useful products and industrial processes. Since ancient times, microbes have been used to produce bread, cheese, wine, and beer. Modern microbial biotechnology has advanced these applications by engineering microorganisms to produce valuable substances such as enzymes, vaccines, insulin, and growth hormones. Microbes are also essential for environmental cleanup, as they can be designed to remove pollutants and industrial waste through biodegradation. Furthermore, biotechnology helps detect harmful microorganisms in food and water, protecting public health and ensuring safety in industries and communities.

Agricultural Biotechnology

Agricultural biotechnology, also called **"ag biotech"**, uses biological techniques to **improve plants**, **animals**, and **food production**. Through genetic engineering, scientists have developed **pest-resistant** and **nutrient-enriched crops** that require fewer chemical pesticides. This helps address global food demands, as experts estimate that food production must increase by about 70 percent by 2050. One important innovation in this field is **molecular pharming**, which uses plants such as tobacco to produce pharmaceutical proteins for medicines. Agricultural biotechnology also supports renewable energy by developing ways to convert crops into biofuels, helping reduce dependence on fossil fuels. However, scientists continue to study its environmental impacts, such as gene transfer and the creation of "superweeds," to ensure responsible use and regulation.

Animal Biotechnology

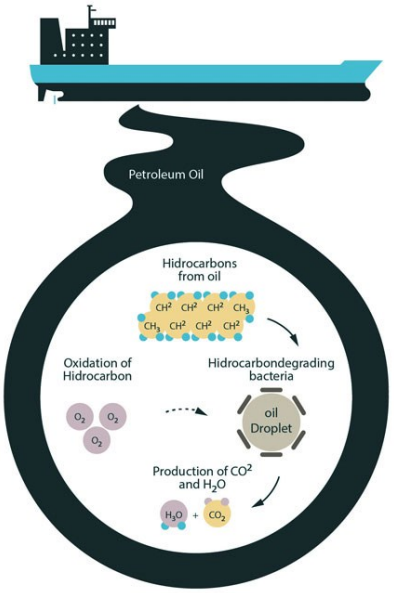


Animal biotechnology involves using animals to produce useful products and to study genetics and disease. Through **genetic modification**, animals like goats, sheep, and cows can be engineered to produce important proteins, such as antibodies or hormones, in their milk. These **transgenic animals** help supply medicines and materials needed for human health. Scientists also perform gene knockout experiments to understand how specific genes affect an organism's functions. A historic milestone in this field was the **cloning** of the sheep **Dolly** in 1997, the first mammal cloned through **nucleus transfer**. Research in cloning and genetic modification has opened new possibilities, such as producing animal organs for human transplantation, but it also raises moral, ethical, and safety questions that require careful consideration.

Forensic Biotechnology

Forensic biotechnology applies DNA technology to **law enforcement** and **criminal investigation**. The most common tool used is DNA **fingerprinting**, which can identify individuals based on their genetic patterns using small samples of blood, hair, or saliva. This powerful method has solved many crimes, proven the innocence of wrongly accused individuals, and identified victims of disasters. Beyond criminal cases, forensic biotechnology is also used to confirm **paternity**, **protect endangered species** by tracking illegal poaching, and trace the sources of disease outbreaks such as *E. coli* or West Nile virus. It even helps verify the authenticity of food and agricultural products to prevent fraud. These applications show how biotechnology promotes justice, safety, and public trust.

Bioremediation



Bioremediation is a special branch of biotechnology that uses living organisms to **clean polluted environments**. It relies on bacteria, fungi, or plants that can break down harmful substances into safer materials. This method is often used to clean up oil spills, industrial waste, and chemical contamination. A famous example is the cleanup of the 1989 **Exxon Valdez oil spill** in Alaska, where scientists encouraged **natural oil-eating bacteria** to grow, speeding up the cleanup process. Similar research was conducted after the 2010 Deepwater Horizon oil spill, where microbial enzymes played an important role in oil degradation. Bioremediation demonstrates how nature, aided by science, can restore balance to ecosystems while reducing the need for harmful chemical treatments.

Aquatic Biotechnology

Aquatic biotechnology focuses on the use of aquatic organisms to **improve food production**, **health**, and **industrial applications**. The most common practice is **aquaculture**, **or fish farming**, which provides nearly half of the world's seafood. Biotechnology has enhanced aquaculture by producing disease-resistant fish and shellfish, such as oysters and salmon, and developing vaccines to reduce losses. Scientists have even created transgenic salmon that grow faster and reach market size in less time, helping meet global food demands. Beyond farming, aquatic biotechnology explores marine life for valuable compounds and materials. Organisms that live in extreme ocean conditions often produce special enzymes and proteins with potential medical or industrial uses. This search for new materials, known as bioprospecting, continues to reveal the ocean's hidden potential as a source of medicines and innovations.

Medical Biotechnology

Medical biotechnology is dedicated to **improving human health** through biological innovations that prevent, diagnose, and treat diseases. Many **modern drugs** and **vaccines** are products of biotechnology, helping millions worldwide. **Genetic discoveries** have allowed scientists to understand illnesses such as cancer, cystic fibrosis, and sickle cell anemia, leading to more effective and personalized treatments. The Human Genome Project mapped all human genes, providing vital information for identifying defective genes and genetic disorders. Projects like the 1000 Genomes Project further this effort by studying genetic variations that influence disease resistance.

Two important innovations in this field are **gene therapy** and **stem cell technology**. Gene therapy works by replacing faulty genes with normal ones, offering the possibility of curing genetic diseases at their source. Stem cells, which can develop into different cell types such as nerve, blood, or muscle cells, are used in regenerative medicine to repair or replace damaged tissues. Although these technologies offer great promise, they also present ethical and legal challenges that society must address responsibly.

BIOTECHNOLOGY REGULATIONS

Because biotechnology deals with **human health** and the **environment**, it is one of the most carefully regulated industries in the world. In the United States, the **Food and Drug Administration** (**FDA**) oversees biotechnology products, while the **Department of Agriculture** and the **Environmental Protection Agency** (**EPA**) also play important roles. Before a product is approved for public use, it must pass strict tests to ensure safety, quality, and effectiveness.

Two key aspects of regulation are **Quality Assurance** (**QA**) and **Quality Control** (**QC**). QA covers all activities that ensure a product meets

safety and performance standards, while QC focuses on laboratory testing during production to maintain consistent quality. These procedures ensure that products such as vaccines and medicines are safe for human use. Regulation also involves patent laws and clinical trials, which evaluate biotechnology products on human volunteers before they are released in the market.

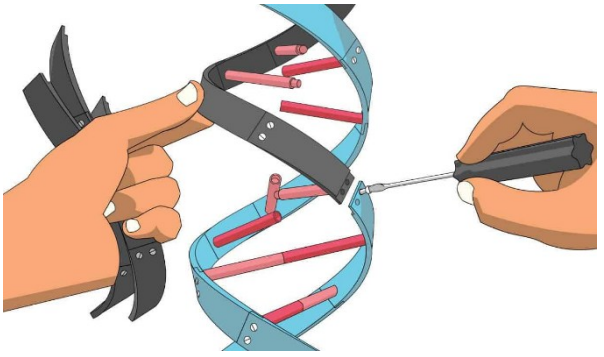
THE BIOTECHNOLOGY "BIG PICTURE"

Although biotechnology is divided into several branches, each area is **interconnected** and contributes to the others. For example, bioremediation uses microbes from microbial biotechnology to clean polluted environments, and medical biotechnology depends on microbes to produce recombinant proteins like insulin. All these areas must also follow the same safety and ethical regulations.

The big picture of biotechnology shows how different scientific fields—genetics, chemistry, microbiology, and engineering—**work together** to solve global challenges. These **collaborations** lead to advances in medicine, agriculture, and environmental conservation. Biotechnology **continues to evolve as a powerful force** that unites science and innovation to improve human life and protect the planet.

BIOLOGICAL CHALLENGES OF THE TWENTY-FIRST CENTURY

Modern biotechnology continues to transform how we **understand** and **treat human diseases**. Many of today's greatest challenges, such as curing life-threatening illnesses, can be overcome through a deeper understanding of biological processes and the creative application of biotechnological tools. The 21st century is often called the biotechnology century because of how genetic and molecular discoveries are reshaping science and medicine.

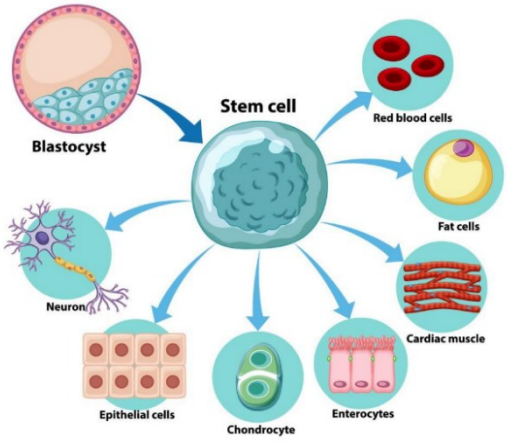


A major breakthrough was the completion of the Human Genome Project in 2003, which **mapped all human genes**. This project revealed that humans have far fewer genes than once expected, yet it provided

scientists with powerful insights into how **genes control life processes**. The project also paved the way for studying the **proteome**—the collection of all proteins made by genes—which helps scientists understand how genes and proteins interact to cause or prevent disease. These discoveries marked the beginning of a new era in personalized medicine, where treatments can be tailored to each person's unique genetic makeup.

One emerging field is **pharmacogenomics**, which uses a patient's genetic profile to design specific and more effective treatments. For example, variations in genes (called single nucleotide **polymorphisms**, or SNPs) can explain why some drugs work well for one person but not for another. In the future, a simple DNA test could help doctors choose the best medication for each patient, reducing side effects and improving recovery. This approach is already being applied in cancer treatment, where gene testing helps doctors select targeted therapies instead of traditional chemotherapy.

Another important advancement is **gene therapy**, which involves replacing or repairing defective genes to treat genetic disorders. Although challenges remain in safely delivering healthy genes to human cells, successful trials have shown promise in treating blood and nervous system diseases. Similarly, **nanotechnology** is being explored to create microscopic tools that can deliver drugs directly to diseased cells, improving precision and reducing damage to healthy tissues.



Stem cell technology also holds great potential for regenerative medicine. Stem cells can develop into different types of cells—such as skin, nerve, or muscle cells—and could one day be used to replace damaged organs or tissues. Scientists are exploring ways to obtain these cells ethically, including from adult tissues or by creating induced pluripotent stem cells (iPSCs) without destroying

embryos. These advances could lead to new treatments for conditions like paralysis, heart disease, and diabetes.

Overall, the future of biotechnology is filled with promise. With continued progress in pharmacogenomics, gene therapy, stem cell research, and nanotechnology, many diseases once thought incurable may eventually be treated or even prevented. The biotechnology century will not only advance medicine but also redefine how humans live healthier and longer lives.